

FUNCTIONAL DOCUMENT

AI-Driven Smart Energy Optimisation and Carbon Analytics Platform

1. Introduction

EN-VISION is an AI-driven enterprise energy intelligence platform designed to monitor, analyse, and optimise energy usage in industrial and enterprise environments. The platform addresses a critical operational gap faced by most organisations: fragmented energy data, delayed reporting, absence of anomaly visibility, no predictive planning, and unclear cost and carbon accountability.

EN-VISION solves this by centralising energy data ingestion, applying AI/ML-based analytics, surfacing anomalies and their root causes, forecasting future demand, and providing KPI-level executive visibility through a role-aware interactive dashboard. The system enables organisations to make data-backed operational decisions to reduce energy waste, control costs, and track sustainability indicators.

2. Product Goal

The overarching goal of EN-VISION is to build a scalable, usable AI-powered platform that enables enterprise-level energy optimisation, cost accountability, and carbon emission management.

Sprint goals for the current review cycle include:

- Complete the authentication and role-based access control stack (JWT + bcrypt, login, session management).
- Deliver profile and change-password account management flows.
- Finalise anomaly intelligence with source attribution (device/zone/department) and dual KPI metrics.
- Standardise cost derivation using India-aligned tariff (INR Rs 8.2/kWh) across all analytics surfaces.
- Integrate smooth route transitions and activation flow for a polished user experience.
- Verify end-to-end build and runtime sanity for frontend and backend integration.

3. Demography (Users, Location)

Users

Target Users:

- Operations managers and energy analysts at industrial or enterprise facilities.
- Sustainability and ESG reporting teams requiring carbon visibility.
- IT/system administrators responsible for user management and platform oversight.
- Executive and management stakeholders needing KPI-level energy and cost dashboards.

User Characteristics:

- Technical proficiency ranges from intermediate (analysts) to non-technical (management viewers).
- Users operate across different access levels depending on their organisational role.
- Primary interaction is through a web-based dashboard with filter-driven navigation.

Location

- The system is designed for deployment within enterprise and industrial environments across India (cost model aligned to Indian average electricity tariff).
- Deployable as an on-premises or cloud-hosted solution accessible via secured web browsers.
- Target geography for initial deployment: Indian enterprises with multi-department or multi-site energy consumption.

4. Business Processes

Energy Data Ingestion

Raw energy consumption data is ingested from connected devices, meters, and departments via CSV upload or direct API integration. The ingestion pipeline normalises data and stores it in structured PostgreSQL tables for downstream analytics.

Energy Analytics

The analytics engine processes ingested energy data to compute KPIs , including total energy consumption, cost breakdowns, trend comparisons, and location/department/device-level consumption views. Filtering by time range, department, and device is supported across all analytics surfaces.

Baseline Modeling

Historical energy consumption is used to compute rolling baselines per device, zone, and department. These baselines serve as reference values for deviation detection and anomaly scoring using residual-based z-score logic.

AI Insights and Anomaly Detection

The anomaly engine identifies deviations from established baselines using residual scoring. Each anomaly is attributed to its source (device, zone, or department) to support root-cause analysis. The system provides dual anomaly KPIs: net anomalies (scope-aware aggregate) and any-device anomaly days (per-device observation count).

Dashboard Visualization

The React/Next.js dashboard presents data across specialised tabs: Home KPI summary, Energy analytics, Anomalies, Forecasting, Carbon, and Power BI. Role-based tab visibility is enforced at the UI level, and animated transitions deliver a polished experience.

Carbon Emission Analytics

Carbon emissions are estimated based on energy consumption data using emission factors. The platform provides carbon visualisation tabs and emission trend tracking to support sustainability reporting.

5. Features

Feature 1: Energy Data Ingestion and Analytics Engine

Description:

The ingestion pipeline accepts energy data from metering systems via CSV upload or API. Ingested data is processed, normalised, and stored in structured database tables (energy_events, daily_energy_summary). The analytics engine computes KPIs, consumption trends, and cost breakdowns in INR based on a standardised tariff of Rs 8.2/kWh.

User Story:

As an Energy Analyst, I want to ingest energy consumption data from our facility meters and view processed KPIs and cost summaries in the dashboard, so that I can identify consumption trends and cost inefficiencies across departments and devices.

Feature 2: AI-Powered Anomaly Detection with Source Attribution

Description:

The anomaly detection engine uses residual-based z-score scoring against computed baselines. Each detected anomaly is attributed to its source (device, zone, or department) with IDs for filter actions. The dashboard presents dual anomaly KPIs (net anomalies and any-device anomaly days), a severity distribution chart, and a clickable anomaly list that supports filter-by-source actions.

User Story:

As an Operations Manager, I want to see anomaly alerts with their source location (device/zone/department) so that I can quickly identify which equipment or area is causing abnormal consumption and take corrective action.

Feature 3: Energy Forecasting

Description:

The forecasting module uses historical consumption patterns to project future energy demand with confidence bands. Forecast data is displayed through a dedicated dashboard tab with trend projections, enabling planning of peak load periods and resource allocation.

User Story:

As a Facility Manager, I want to view projected energy consumption for upcoming periods so that I can plan for peak demand, avoid cost overruns, and improve operational scheduling.

Feature 4: Carbon Emission Analytics

Description:

The carbon analytics module calculates and visualizes CO2 equivalent emissions based on energy consumption data. The dashboard provides emission trend views, source-level carbon contribution breakdown, and sustainability KPIs to support environmental reporting and ESG compliance.

User Story:

As a Sustainability Officer, I want to track our facility's carbon emissions over time and by department, so that I can prepare accurate sustainability reports and identify areas for emissions reduction.

Feature 5: Role-Based Access Control and Account Management

Description:

The platform implements JWT + bcrypt-based authentication with three distinct roles: Admin, User, and Viewer. Role-based sidebar controls restrict tab access dynamically. Account management includes profile update and password change flows accessible through the header navigation.

User Story:

As a System Administrator, I want to assign roles to platform users so that each user sees only the analytics tabs and actions relevant to their responsibilities, ensuring data security and access control.

6. Authorisation Matrix

Role	Home/KPI	Energy Analytics	Anomaly Detection	Forecasting	Carbon Analytics		Power BI	Ingestion Admin	User Management
Admin		Full Access	Full Access	Full Access	Full Access	Full Access	Full Access	Full Access	Full Access
Energy Analyst (User)		Full Access	Full Access	Full Access	Full Access	Full Access	No Access	No Access	No Access
Viewer		Read Only	Read Only	Read Only	No Access	No Access	No Access	No Access	No Access
Organization User		Full Access	Full Access	Full Access	Full Access	Read Only	No Access	No Access	No Access

7. Assumptions

- The platform is developed and tested by a two-member student team, with both members contributing to backend (FastAPI/Python) and frontend (Next.js/React) development equally.
- Energy consumption data is assumed to be available in structured CSV format or via pre-configured API endpoints for ingestion. Real-time IoT device integration is out of scope for the current sprint.
- The system is deployed in a local development environment using PostgreSQL as the primary database, with production deployment assumed on a cloud-hosted or on-premises Linux server.
- The cost model uses a fixed Indian electricity tariff of Rs 8.2 per kWh as the baseline. Configurable tariff per site is planned for a future sprint.
- Users accessing the platform are assumed to have a modern web browser (Chrome, Edge, Firefox) and stable internet/intranet connectivity.
- Authentication token management is handled via browser-side storage/cookie; no third-party identity provider (SSO/OAuth) integration is in scope for the current sprint.